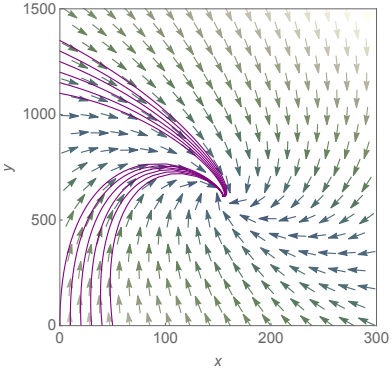
# MAST20030 Differential Equations

The University of Melbourne
School of Mathematics and Statistics

Semester 2, 2025

Luke G. Bennetts

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These notes are for the use of students of the University of Melbourne enrolled in MAST20030 Differential Equations.

Original notes by Jesse F. Collis and Antoinette Tordesillas

# Meet the MAST20030 teachers

Prof Luke Bennetts
Lecturer

- Lectures
- Consulting hours
- Canvas/LMS materials
- Assessments

Dr Chris Baker
Co-teacher

- Tutorial management
- Assessments

Mr Howie Zhou
Course Assistant

- Ed Discussion
- Problem booklets
- Also a Tutor

# Welcome to MAST20030 Differential Equations

- Canvas/LMS contains all info on subject
- Please read the short welcome page on Canvas/LMS, then the Meet the Teaching Team and Communicating with Staff & Peers pages

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## Assessment

- Two assignments [20%], due in Weeks 5 and 12 and released two weeks prior
- A 45-minute mid-semester test in Week 7 [15%]
- A three-hour exam during the examination period [65%].

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## Lectures

- Come to the lectures to engage with your peers and Luke, to listen, to ask questions and to create the positive environment that university campuses are all about!
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- **Mathematical Physics:** string theory; quantum physics; general relativity including black holes; phase transitions; polymers; ...

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In UoM Sciences:

- imaging cells; gravitational waves; ocean mixing; global warming; earthquakes; ...

Across the university:

- self-driving cars; renewable energy; nano medicine; conservation ecology; ...

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**Case studies**

We will use *case studies* to learn how the methods we cover in lectures are applied to solve real-world problems.

# Course outline: Six topics

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

# Topic 1: Linear ODEs and IVPs

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1. How linear ODEs arise in real-world applications
2. Four methods for solving linear ODEs and IVPs
3. Connect ODEs to linear algebra
   - From Handbook: "Explain how linear algebra dictates the structure of the solution space of a linear differential equation"

# Lecture 1

We will:

1. Define and classify ODEs
2. Define IVPs
3. Revise the integrating factors solution method

## Poll Everywhere

`PollEv.com/lukebennetts886`

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# Linear ODEs

## Definition (Classification of ODEs)

A *linear ODE* (ordinary differential equation) is a linear combination of derivatives of an unknown function $y(t)$ whose coefficients may be functions of $t$ but not $y$:

$$f_n(t)y^{(n)}(t) + \ldots + f_1(t)y'(t) + f_0(t)y(t) = g(t)$$

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- If $g(t) = 0$ then the ODE is *homogeneous*.
- If $g(t) \neq 0$ then the ODE is *inhomogeneous*.

# **Exercise 1.1**: Classify the following ODEs

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   This is a boundary value problem, which we will meet later

# Integrating factor (revision)

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$$\frac{1}{t}y(t) = \int \frac{1}{t}\mathrm{d}t = \log t + c \qquad \Rightarrow \qquad y(t) = t\log t + Ct$$